Creating and Delivering Operational Plans

Lesson 5

WESTON COLLEGE | Mindful Education

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Study Information and Tips

This Study Text replicates the video content for the lesson

Supplementary knowledge can also be found in the information sheets and text boxes that accompany the videos, as well as in the resources in the Insights section

Activities to help you consolidate your knowledge are deliberately delivered through practice questions and exercises found in the Test and Apply sections on the VLE. These help you to further build on your knowledge and apply the learning in a way that supports your assessment

You can use this Study Text to test your understanding of the learning content at the end of each lesson. It may also be helpful to refer to it when preparing for your assessments

We recommend that you also produce your own notes in addition to the Study Text. Research has shown that handwritten notes can be very effective for learning retention

If you decide to print this Study Text, set the layout to print 4 or 6 slides per page to save on paper and ink

Problem-solving and Decision-making

Common problem-solving approaches

Analytical tools to support effective decision-making

Approaches to Problem-solving

Even the most well-considered plan can run into problems in its implementation. If these escalate, they can result in unexpected downtime, safety errors or lower team morale. Being able to resolve problems – and to keep them from recurring – is therefore critical for business.

We'll now look at one popular approach to problem-solving: root cause analysis.

Root Cause Analysis

This is not a single methodology but a collection of tools to uncover the underlying reasons behind why a problem exists. It's based on the principle that problems can best be solved by correcting their root causes, rather than addressing the symptoms. Instead of 'putting out fires', it looks for ways to prevent them.

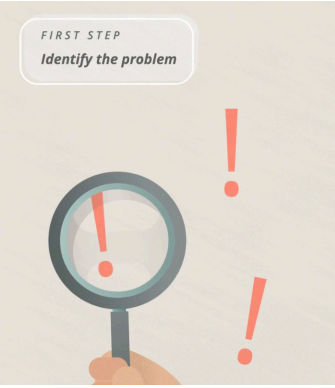
The approach originated in the Japanese car industry as a way to reduce production inefficiencies; it's now widely used to investigate all sorts of problems, from air traffic accidents to patient safety issues.

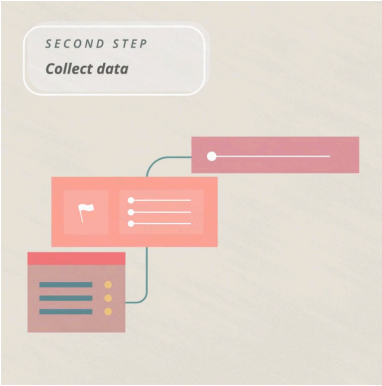
Root Cause Analysis (cont.)

The first step in the process is to identify the problem. For a manager, this means recording what is happening – the problem's visible symptoms.

The next step is to collect data. The aim here is to establish the extent of the problem, to determine how long it has existed, and assess the impact it's having.

With this evidence in hand, it's possible to move on to the third step, which is to identify as many causal factors for the situation as possible. The goal is to find one or more root causes of the problem – the whys behind it. Without this understanding, a problem may be remedied in the short term, but not truly resolved.





The Five Whys

One popular method to uncover the root cause is known as the Five Whys technique.

This is a systematic way to drill down to the underlying reasons for a problem, and show the relationships between them.

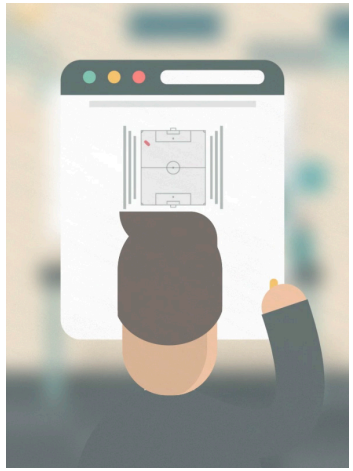
The technique requires the question, ‘why?’ to be asked, ideally five times.

The premise is that each question will lead into a fresh ‘why?’, until the answer to the problem reveals itself.

Let’s see how this might work in practice.

The Five Whys – Example

Brian is a manager in the parks department of the local council. Whilst monitoring the work of his team, he's noticed that some local pitches are not ready for the start of the football season. Before he jumps to conclusions, he gets the team together to find out more.



Brian asks the team members why the pitches have not been prepared.

They tell him that they're behind schedule because they are sharing one grass mower. Brian now wants to know why this is the case.

The team members explain that several of the mowers aren't functioning. Brian again asks why.

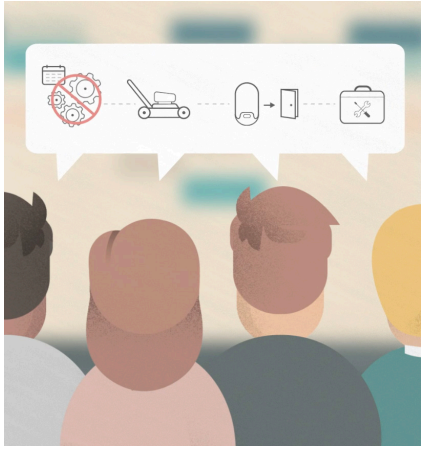
The Five Whys – Example (cont.)

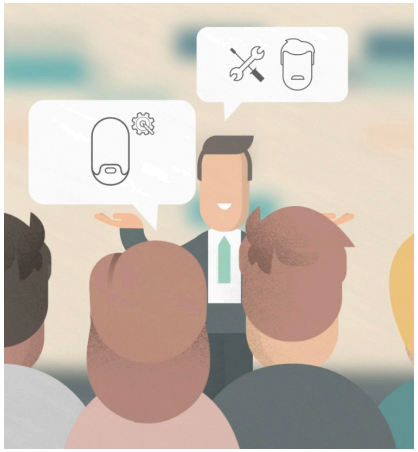
The team members reply that the monthly servicing of equipment wasn't performed.

Brian follows up with another why and the employees clarify that it's because their teammate Charlie has left, and servicing the mowers was his job.

Brian asks why the equipment couldn't be repaired by someone else.

The team members admit Charlie was the only person trained to repair the mowers. This is the root cause of the problem.





Process, Policy or Person

While advocates of the method recommend asking ‘why’ five times, depending on the problem it may take fewer or significantly more questions to get to the actual root.

Some experienced practitioners say that the root cause will be found when the answer given is a ‘process, policy or person.’ This is reflected in our example, where the absence of one person created a logistical problem for the whole team.

Once sufficient data has been collected and the underlying reason or reasons for the problem have been pinpointed, the fourth step is to recommend and implement solutions.

An Iterative Process

In our example, the root cause is that only one team member is trained to fix the mower. Brian needs to arrange for more training for his team. At the same time, he'll set up a new servicing plan so that future breakdowns don't have such dramatic effects.

The ultimate goal of root cause analysis is to find a solution that will prevent the problem from happening again. This will require further analysis of how the solution should be implemented, who will be responsible for it, and any associated risks. All of these aspects will need to be considered in Brian's situation.

It may be difficult for a single analysis to prevent the recurrence of a problem, which is why it's often considered to be an 'iterative' process. With more experimentation, a permanent solution may be reached over time.

Brainstorming

Some problem-solving techniques can be quite creative and visually engaging. We'll now look at three such approaches: brainstorming, mind mapping and the fishbone, or Ishikawa, diagram.

Brainstorming is 'a group problem-solving technique that involves the spontaneous contribution of ideas from all members of the group.' It's often used by teams to resolve an issue that requires collective input.

Brainstorming (cont.)

The process starts with participants being asked to generate as many ideas to resolve the issue as possible.

At this point, all ideas are accepted without judgment, no matter how outlandish they might first appear.

Once all the suggestions are on the table, but not before, each idea is evaluated against the others. The hope is that this type of 'free thinking environment' will unlock the group's creativity, so they can identify the root cause of the problem and ways to resolve it.

Brainstorming (cont.)

Brainstorming is based on the premise that people working collectively to find a solution will be more productive and innovative than each person working individually. As everyone is invited to participate, it can improve team cohesion and lead to increased self confidence.

However, brainstorming must be carefully facilitated to ensure equal participation by all parties, and it can take some time to produce results.

Mind-mapping

Another approach which shares some elements with brainstorming is known as mind-mapping. This is a ‘graphical technique for visualising connections between several ideas or pieces of information.’

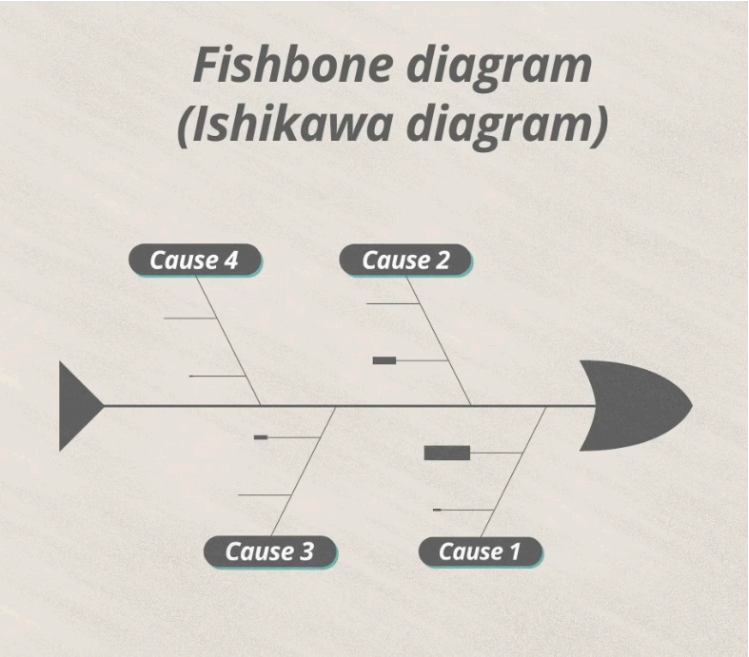
By showing all the aspects of a problem and how they are connected, it enables a manager to organise their thoughts and literally see the bigger picture.

There are no formal rules for creating a mind map, although some people like to illustrate it with different shapes and colours to stimulate ideas.

Fishbone Diagram

The final creative tool is the ‘fishbone’ diagram – so named because its shape resembles a fish skeleton. Although it dates to the early 20th century, the method was made popular Japanese quality management expert Kaoru Ishikawa in the 1960s. You’ll see it referred to as either the fishbone or Ishikawa diagram.

This is another, more visual approach to root cause analysis. The diagram maps ‘cause and effect’ to identify possible reasons why a problem occurred, and discover its root cause. In this way it’s similar to the 5 Whys technique, but also features aspects of brainstorming and mind-mapping.



Fishbone Diagram (cont.)

Because of its Japanese origins, the traditional diagram starts with the problem on the right, and builds across the page to the left. However, it's equally effective presented the other way around.

Whichever direction it starts from, the analysis proceeds from the 'head' of the fish. Here the manager will write down the exact nature of the problem. This might include who is involved in it, and where it is located – for example, within the department as a whole or in a particular team.

Next comes the brainstorming of potential factors that could be causing the problem. These are usually arranged in categories.

Fishbone Diagram (cont.)

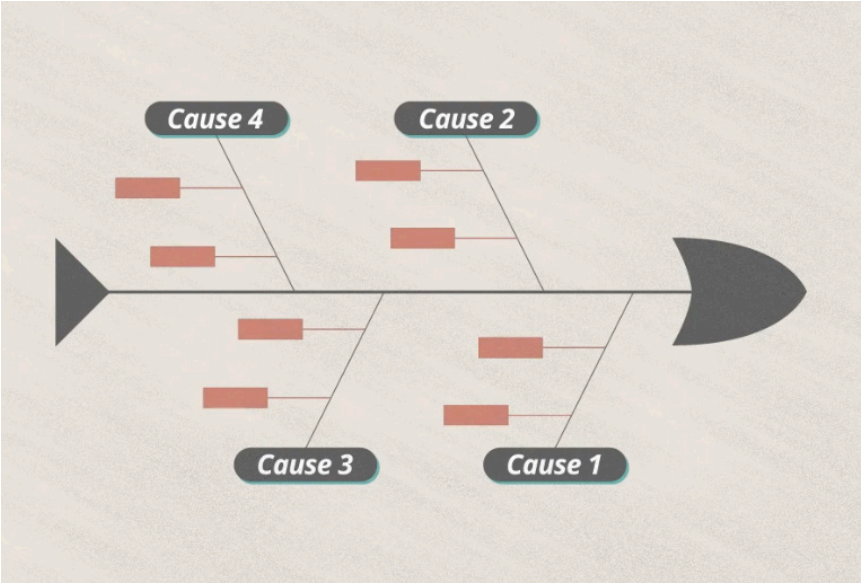
In a manufacturing setting, which is where the model originated, the categories might include machinery, materials, methods and measurement. In a healthcare context, on the other hand, it may be more relevant to include people, process, supplies and environment.

The selected categories are then added to the diagram as ‘ribs’, extending out from the central spine.

Fishbone Diagram (cont.)

Finally, the manager will explore the possible causes for each of these factors. These are represented by new ‘bones’ branching out from the existing ones.

The finished diagram could look something like this...



Fishbone Diagram (cont.)

The Ishikawa diagram can give a visual account of the cause and effect relationship in a problem. It allows managers to see all the causes in an organised way, and to check if the root cause is found multiple times. This would highlight that the underlying reasons for the problem are further reaching than first anticipated.

However, critics argue that on the page, each cause may look equally important, meaning time is wasted on causes that have little effect on the problem. And complex problems may produce diagrams with many branches, leading to confusion and some dead ends.

Whichever method is used, once a problem has been identified and analysed, the manager will need to weigh the options and make a decision on how to address it.

Decision-making

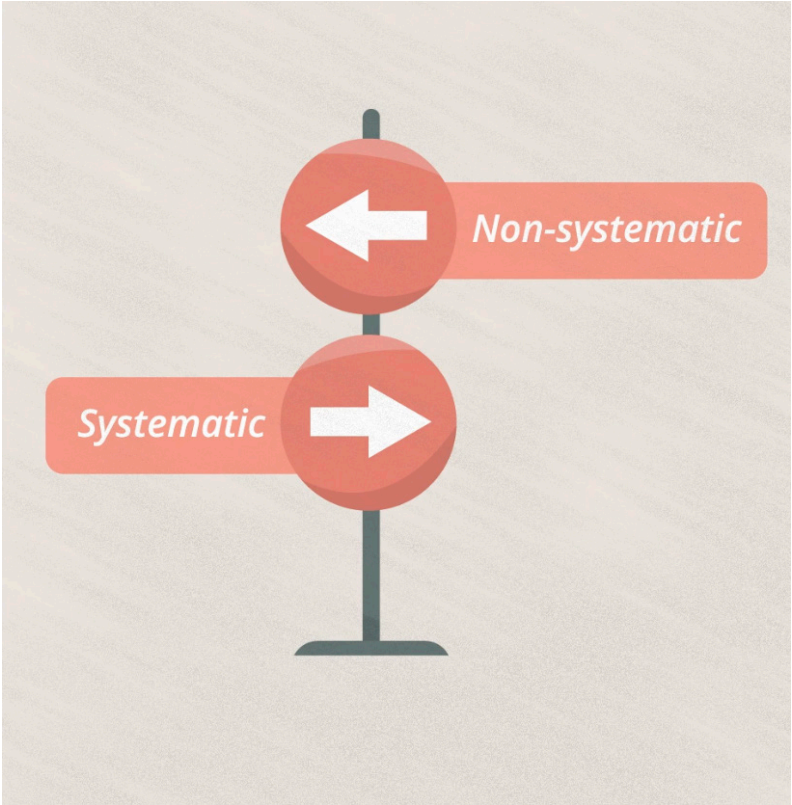
While analysis is an important part of problem-solving, to actually tackle the problem a manager will usually need to make a decision, or a series of decisions.

In fact, decision-making is part of an operational plan from its inception. Managers need to decide how they will measure performance, assign work and allocate resources. The process will continue right through the implementation period, where they will make decisions on how to handle problems uncovered in monitoring. The outcomes of this process will then feed into the next plan.

Non-systematic Decision-making

Decision-making approaches may be loosely grouped into two main categories: non-systematic and systematic.

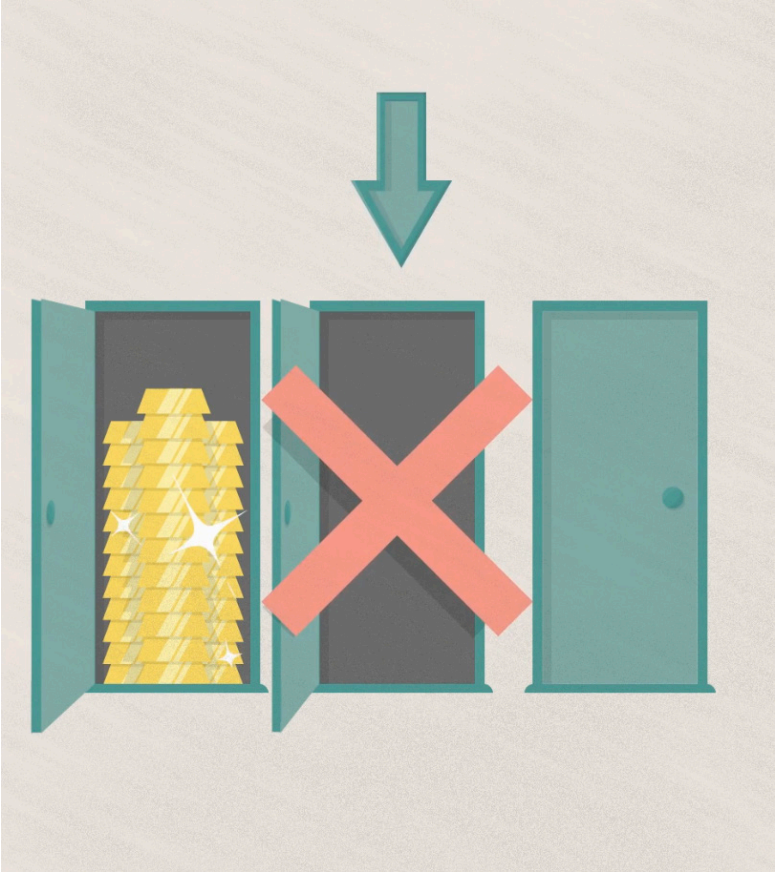
Non-systematic approaches are quick and intuitive. They don't involve lengthy analysis – in fact, they may not even be consciously set out. If you look at the decisions you make over the course of a working day, you'll find that many of them fall into this category.



Non-systematic Decision-making (cont.)

One non-systematic option is to follow the path of others who were in a similar situation. Such an approach benefits from the experience of colleagues, and can help speed up the decision.

However, it also means missed opportunities to try untested, but possibly better options. In addition, what has worked before may not work again in a different context.



Non-systematic Decision-making (cont.)

A non-systematic choice may also be based on our own experience. This has the advantage of familiarity, and may lead to better decisions on issues we've encountered before. However, it's also possible to have overconfidence in our expertise, remember things inaccurately, or fail to detect our own biases. All of these may lead to the wrong choice and poor results.

Finally, a non-systematic decision may rely purely on 'intuition.' Intuition is the ability to understand something using our instincts, rather than conscious reasoning. This can be a powerful shortcut to decision-making. But it can also be subject to personal bias.

Systematic Decision-making

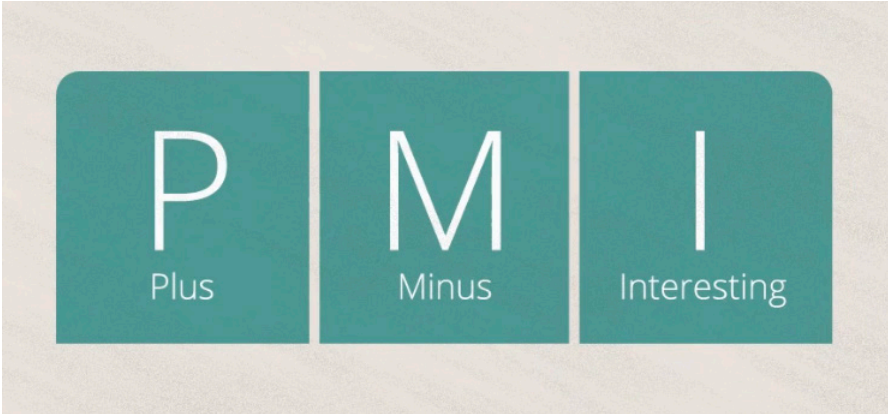
To minimise the impact of bias, as well as other drawbacks of these approaches, it helps to use systematic techniques.

A ‘systematic’ decision-making approach uses a logical and orderly process that has been developed in advance. This kind of approach is less likely to miss critical factors, which can especially help in situations where the choice is complex.

We’ll look at two systematic approaches here.

The PMI Model

The Plus, Minus and Interesting, or PMI model was created by business psychologist Edward de Bono. It divides actions by whether their outcomes are positive, negative, or neutral – but in some way interesting and relevant – to the choice.



So, for example, the manager of a library may be contemplating whether to introduce a paid reservation system for new books.

The PMI Model (cont.)

The ‘pluses’ here would be more revenue for the library, enabling it to expand its offering.

The ‘minuses’ would include the disadvantage to less affluent library patrons – they might be discouraged from borrowing new books.

And in the neutral but ‘interesting and relevant’ category there is the manager’s belief that patrons who pay to reserve books will be more likely to pick up and read them.

The PMI Model (cont.)

PMI is easy to use by individuals or groups and can be applied quickly. However, the simplicity of the model can also be a weakness. The structure is so loose that it can reinforce existing preferences instead of testing them.

For example, in PMI there's no requirement to consider the wider objectives as a factor in the decision – but these are important to ensure the outputs from the model are fit for purpose.

In the case of the library, there is an overriding objective to serve all patrons equally – and this should outweigh the extra revenue the library can generate. Without taking this objective into account, however, there's a danger that the pluses of higher revenue would overcome the minus of lower access for patrons.

Grid Analysis

Our second systematic approach is Grid Analysis.

This is a useful method when the decision involves a number of options, with multiple factors to consider. Using a grid or table, the options are set out in rows. The factors affecting the options are entered in columns, and assigned a numerical value based on their importance.

Each option is then ranked on how well it scores on each factor. Finally, these rankings are multiplied by the value of the factor to get a total score for each option.

Grid Analysis (cont.)

So, for example, a manager planning an away day for the team could identify three possible venues as options, and rank these against factors like hire costs, travel time and facilities. The manager would then select the highest scoring venue.

By quantifying options and taking all the factors into account, Grid Analysis can give a precise view of the choice at stake. But not all choices can be easily quantified in this way. And as we've seen, many decisions are made without conscious analysis.

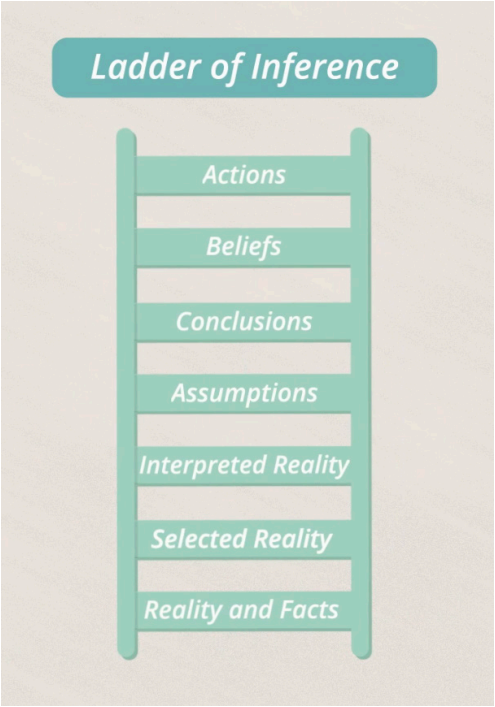
Factors	Hire costs	Travel time	Facilities	Total
Venue A	6	4	8	18
Venue B	2	6	3	11
Venue C	7	5	8	20

The Ladder of Inference

Decisions should be made on a solid foundation of evidence. When they're based on weak or insufficient data, they tend to lean heavily on non-systematic approaches like intuition. As we have already seen, this can be a problem because personal biases can distort events or interactions, and often lead us to the wrong conclusion.

A model known as The Ladder of Inference can encourage the decision-maker to resist this tendency, and stay focused on the evidence.

The model was first developed in 1970 by Harvard professor Chris Argyris and made popular by learning specialist Peter Senge. Using the metaphor of a ladder, it follows the thought process that people undertake, usually unconsciously, when making a decision. This process consists of seven steps, which start with the facts of the situation, and end with a decision.



The Ladder of Inference (cont.)

According to the model, people are first confronted with the Reality and Facts of a situation.

However, when they contemplate these facts, they unconsciously narrow in on the things that matter most to them. This is their Selected Reality.

Next they give the selected facts a personal meaning. This creates an Interpreted Reality. Based on this interpretation they make Assumptions – sometimes without deeper consideration.

They then draw Conclusions from the assumptions. The conclusions are integrated into their existing system of Beliefs.

And finally, these beliefs form the justification for their Actions.

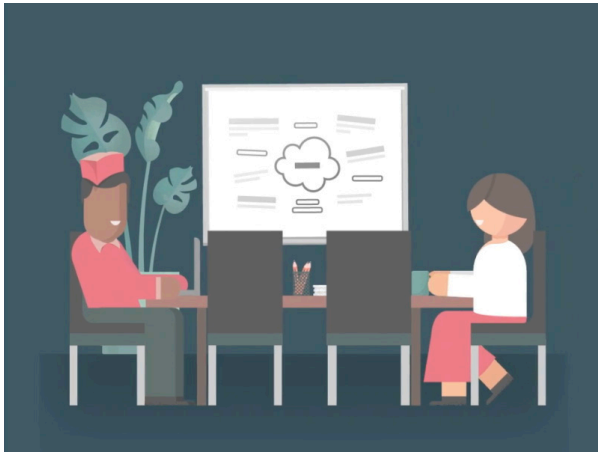
The Ladder of Inference – Example

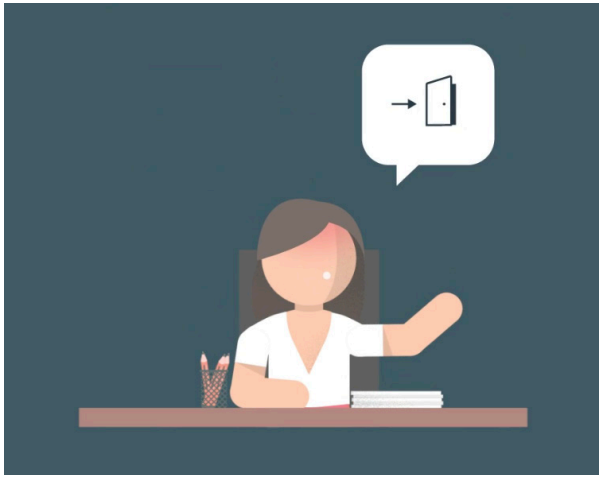
While this process happens very quickly, the model can help break it down, and help us to examine it in slow motion, to assess our reasoning for a decision before we make it.

Let's see how this might work in practice.

Clara is the operational manager at a multiplex cinema. One of her team members, Louis, has been helping her plan a forthcoming film festival.

However, she has discovered that he is seriously behind schedule. Although they've met briefly to discuss the issue, Louis had little to say and was in a rush to leave.





The Ladder of Inference – Example (cont.)

This has created a dilemma for Clara, as the delay will impact on her own tight deadlines. She's wondering if Louis is actually capable of doing the job, or whether she should ask him to step down from the project.

To test her reasoning, Clara retraces her thought process using the Ladder of Inference.

The starting points are the Reality and Facts that form the situation we've just set out – Clara has records of Louis' work, and a good sense of where he has fallen short.

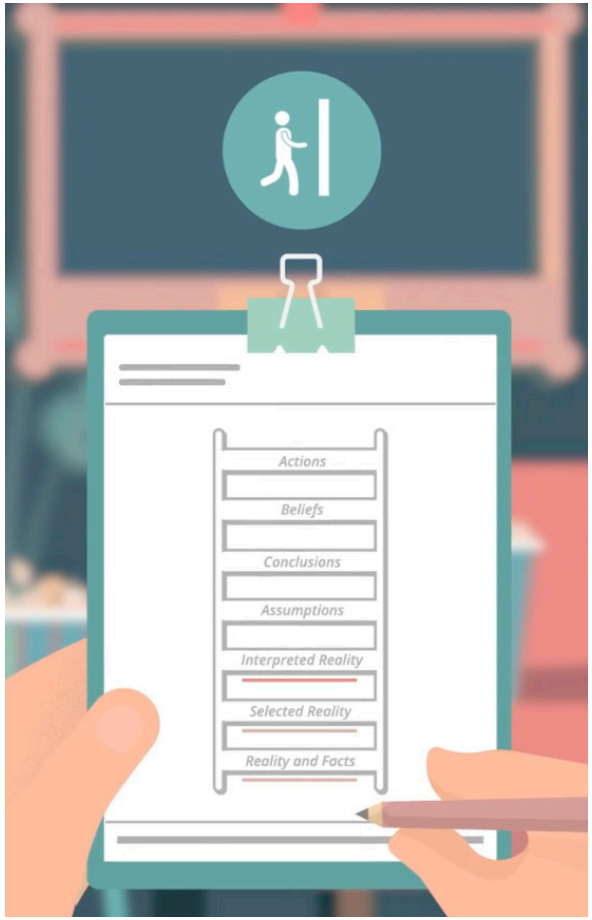
The Ladder of Inference – Example (cont.)

The next step is Selected Reality.

In their meeting, Clara focused on the fact that Louis didn't tell her he was behind schedule, and that he was in a rush to go.

Third is Interpreted Reality.

Clara recognised Louis' avoidance behaviour – she had seen it before, usually in people who want to evade their responsibilities. So she interpreted it in this context.



The Ladder of Inference – Example (cont.)

The fourth step is Assumptions.

As a result of the meeting, Clara assumed that Louis is either no longer committed to the film festival, or is not capable of doing the job.

This is followed by Conclusions.

Clara concluded that she could no longer trust Louis to complete his remaining tasks on schedule. And as he offered no explanation, she surmised that he doesn't want to fix the problem.

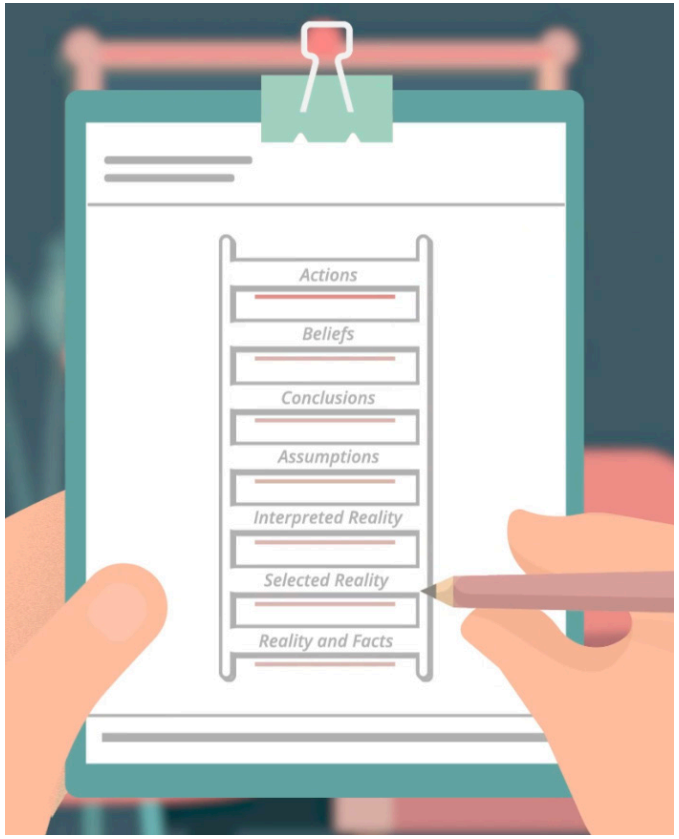
The Ladder of Inference – Example (cont.)

This leads to the sixth step – Beliefs.

Clara's previous beliefs about 'people like Louis' have made it easier to believe that he is a liability to the operation.

Finally, there is the seventh step – the Actions. These come about as a result of all the other steps on the ladder.

Because of her beliefs about Louis, Clara is poised to replace him on the project. However, as she examines the path to this decision, Clara realises that her evidence for the choice is slim.



The Ladder of Inference – Example (cont.)

What if Louis was committed to planning the film festival, but struggling with a task, and afraid to ask for help? Maybe he had some serious personal issue which affected his work, but which he was reluctant to share? Clara reconsiders the situation, and decides to have a longer one-to-one with Louis. Perhaps arranging a PDP might get him back on track.

Professor Argyris suggests that if we don't think about the inferences that inform our decisions, a vicious cycle of assumptions and conclusions will take over, so that the next time a similar choice presents itself, we may ignore certain facts altogether.

However, if we become aware of why we take certain actions, both our relations with the team, and our operational plans will benefit.

Evidence-based Decision-making

As we've seen, decisions should be based on a solid foundation of evidence. For this reason, 'evidence-based practice' has risen to prominence since the late 2010s. It has gained significant traction in various fields, including healthcare, education and management.

Find out more about how it's used in decision-making in this lesson's associated information sheet.

Recap

In this lesson, you have learned about:

- Common problem-solving approaches
- Analytical tools to support effective decision-making